IOT AND ROBOTICS

Department of Electronics &Communication Engineering

Report Name : Home Automation System using HC 05 Bluetooth Module.

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Home Automation System using HC-05 Bluetooth Module

Objective:

To design and implement a low-cost, Bluetooth-controlled home automation system that enables users to operate household appliances wirelessly using a smart phone. The system uses the HC-05 Bluetooth module to receive commands and an Arduino microcontroller to control electrical devices.

Components Required:

Component	Quantity
Arduino UNO	1
HC-05 Bluetooth Module	1
1-Channel Relay Module	1
Android Smartphone	1
230V AC Appliances (Bulb)	1
Jumper Wires	several
Power Supply (5V)	1

Circuit Design:

Here is the wiring configuration:

HC-05 Bluetooth Module:

VCC → 5V (Arduino)

GND → GND (Arduino)

TX → RX (Pin 1 of Arduino via voltage divider)

RX → TX (Pin 0 of Arduino)

Relay Module:

IN1 → Arduino Pin 8

VCC → 5V

GND → GND

Arduino Connections:

Pin 1 → HC-05 TX

Pin 0 → HC-05 RX (with voltage divider)

Note: Always use a voltage divider or level shifter on HC-05 RX to avoid damage.

Code (Arduino Sketch):

```
char command; // Variable to store incoming Bluetooth command
int relayPin = 8; // Relay module control pin

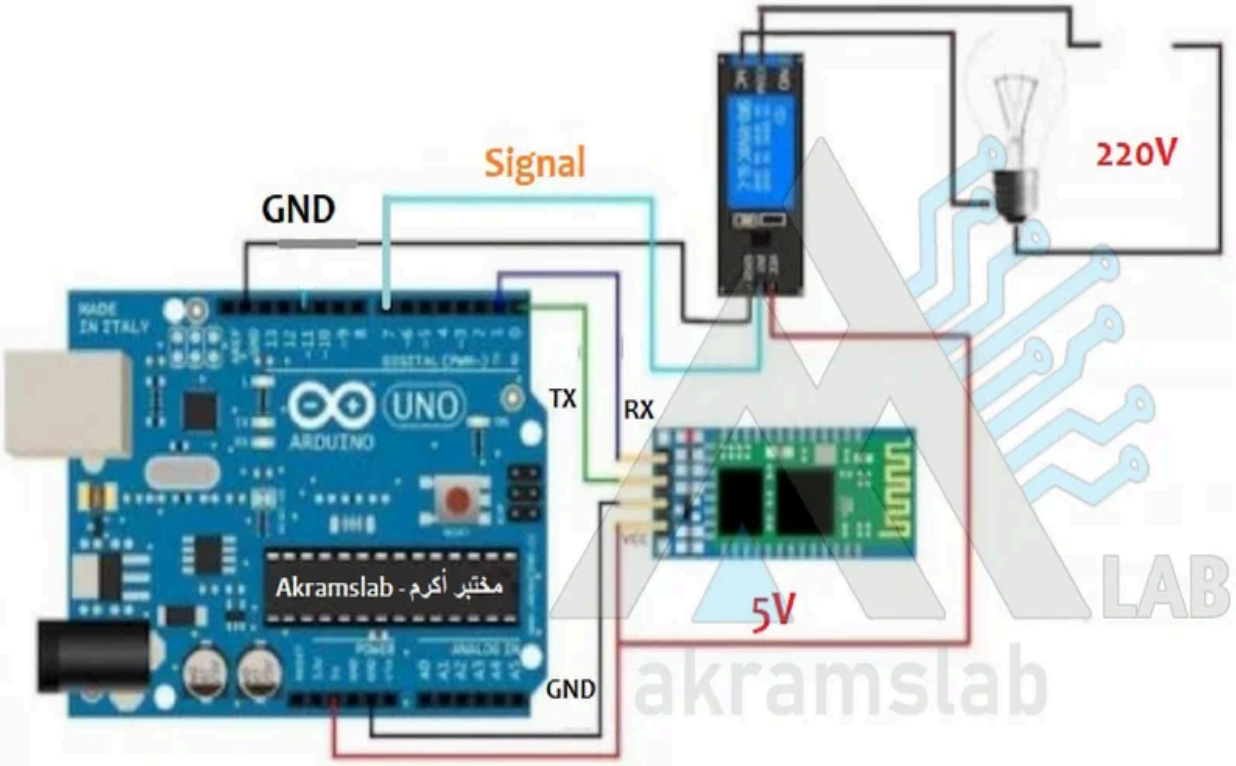
void setup() {
  Serial.begin(9600); // Start serial communication
  pinMode(relayPin, OUTPUT); // Set relay pin as output
  digitalWrite(relayPin, LOW); // Keep relay OFF initially
  Serial.println("Bluetooth Relay Control Ready");
}
```

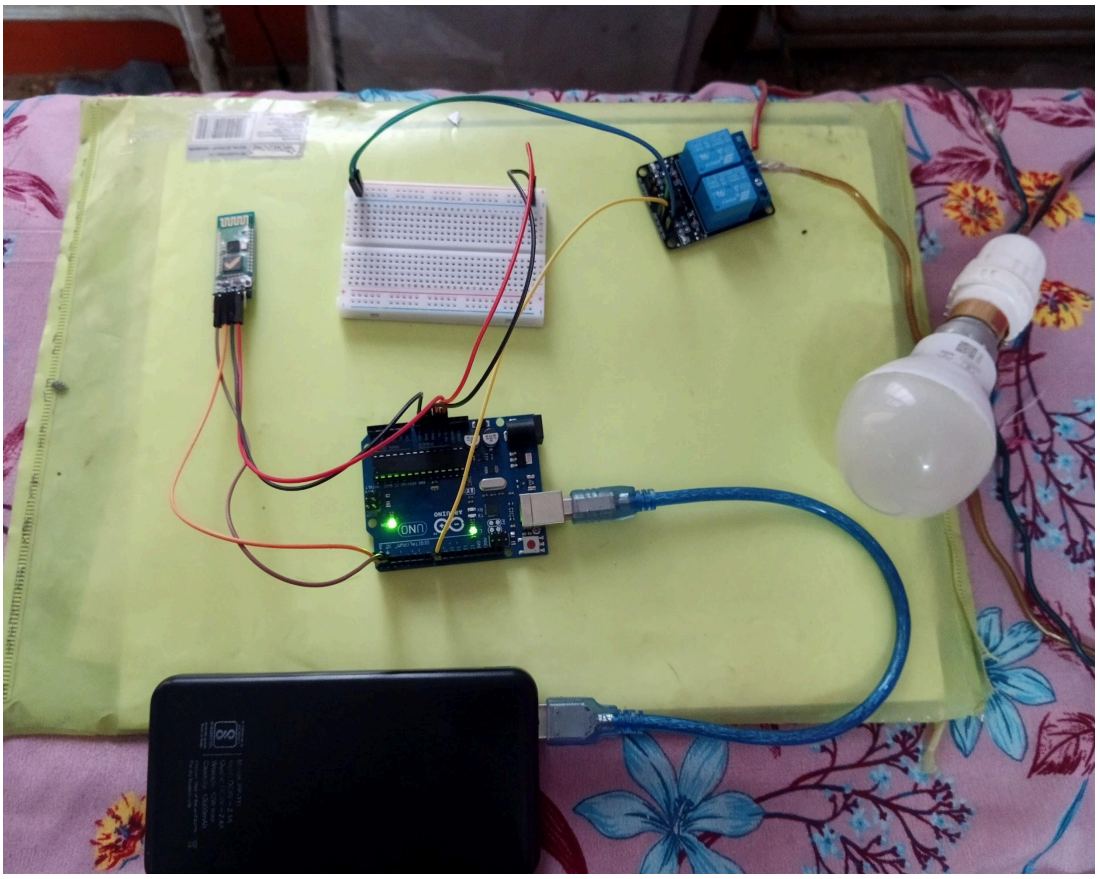
```
void loop() {
  // Check if data is available from HC-05
  if (Serial.available()) {
    command = Serial.read();

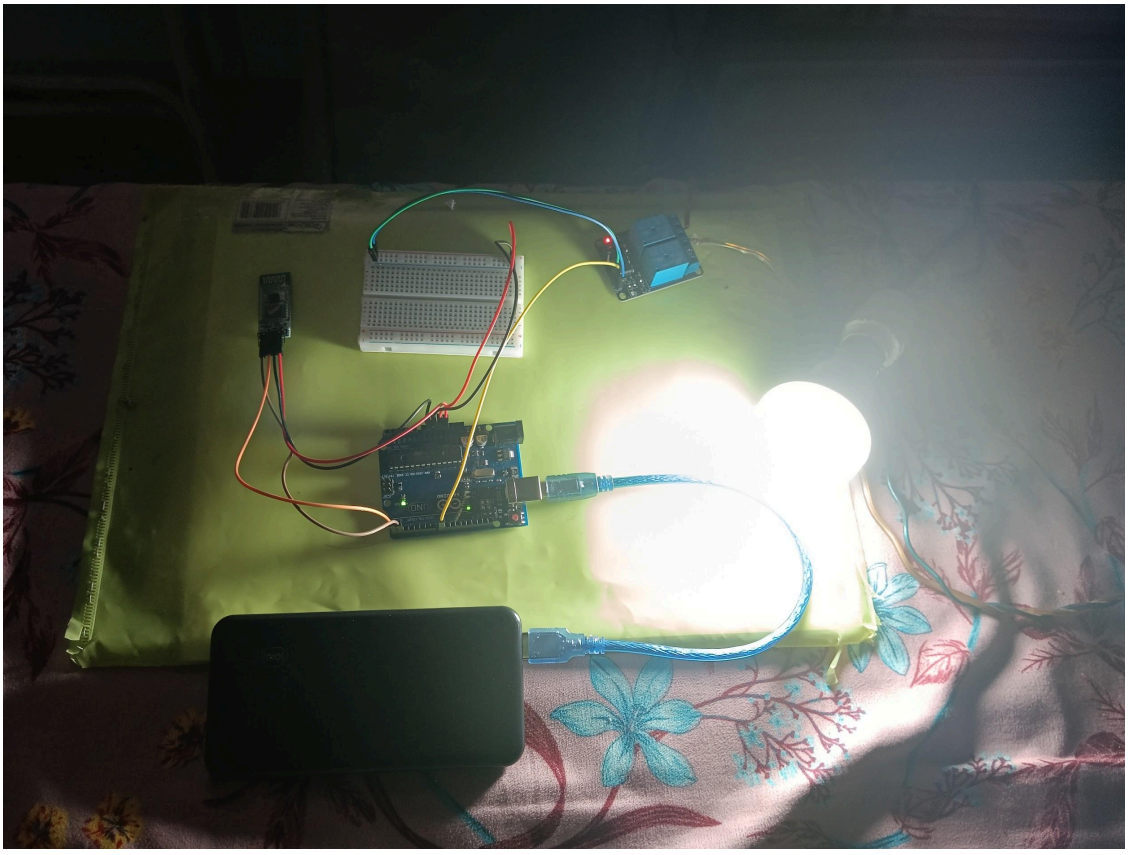
    // Turn ON bulb
    if (command == '1') {
      digitalWrite(relayPin, HIGH); // Relay ON
      Serial.println("BULB ON");
    }

    // Turn OFF bulb
    else if (command == '0') {
      digitalWrite(relayPin, LOW); // Relay OFF
      Serial.println("BULB OFF");
    }
  }
}
```

Circuit Diagram :


Signal
GND
220V
TX
RX
5V
GND
UNO
ARDUINO
Akramslab - مختبر أكرم
akramslab
LAB





Testing Logs:

Test Case	Command Sent	Expected Output	Result
1	Relay	ON	Passed
2	Relay	OFF	Passed
3	Invalid char	No changed in output	Handled

Note : All tests were conducted using a basic Android Bluetooth terminal app (like “Bluetooth Terminal HC-05”).

Conclusion:

This project successfully demonstrates a functional home automation system using the HC-05 Bluetooth module and Arduino UNO. It allows remote switching of household appliances via Bluetooth, offering a cost-effective, scalable, and user-friendly solution for smart home applications.

Future improvements :

it may include:

Adding feedback/status monitoring.

Integrating with Wi-Fi for cloud control.

Voice control via Google Assistant.